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SIMATS ENGINEERING **SIMATS Online Programming Lab** **9 NAGA TEJASWITHA 192371057RD**

JAVA PROGRAMMING

QUESTIONS

- Q1** Palindrome number/string 10 marks
- Q2** First n perfect numbers 10 marks
- Q3** Decimal number equivalent to binary number and octal number 10 marks
- Q4** Nth max number and nth min number in array then sum of it and difference of it 10 marks
- Q5** Find the average positive and negative numbers upto -1 10 marks
- Q6** Removing duplicates in array 10 marks

Q1 Palindrome number/string 10 marks

Palindrome number/string

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int num = sc.nextInt();
6         int temp = num;
7         int rev = 0;
8         while (num > 0) {
9             int rem = num % 10;
10            rev = rev * 10 + rem;
11            num = num / 10;
12        }
13        if (temp == rev)
14            System.out.println("Palindrome");
15        else
16            System.out.println("Not Palindrome");
17    }
18 }
```

Input

121

Output

Palindrome

Run **Save**

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- Q6** Removing duplicates in array 10 marks
- Q7**

Q2 First n perfect numbers 10 marks

First n perfect numbers

Your Code

```
1 import java.util.*;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int count = 0;
7         int num = 2;
8         while(count < n) {
9             int sum = 0;
10            for(int i = 1; i <= num/2; i++) {
11                if(num % i == 0)
12                    sum += i;
13            }
14            if(sum == num) {
15                System.out.println(num);
16                count++;
17            }
18            num++;
19        }
20    }
21 }
```

Input

3

Output

6
28
496

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QUESTIONS

Q1
Palindrome number/string
10 marks

Q2
First n perfect numbers
10 marks

Q3
Decimal number equivalent to
binary number and octal
number
10 marks

Q4
Nth max number and nth min
number in array then sum of it
and difference of it
10 marks

Q5
Find the average positive and
negative numbers upto -1
10 marks

Q6
Removing duplicates in array
10 marks

Q3 Decimal number equivalent to binary number and octal number 10 marks

Decimal number equivalent to binary number and octal number

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String binary = sc.next();
6         String octal = sc.next();
7         int decimal1 = Integer.parseInt(binary, 2);
8         int decimal2 = Integer.parseInt(octal, 8);
9         System.out.println("Binary to Decimal = " + decimal1);
10        System.out.println("Octal to Decimal = " + decimal2);
11    }
12 }
```

Run

Save

Input

1010
17

Output

Binary to Decimal = 10
Octal to Decimal = 15

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QUESTIONS

Q1
Palindrome number/string
10 marks

Q2
First n perfect numbers
10 marks

Q3
Decimal number equivalent to
binary number and octal
number
10 marks

Q4
Nth max number and nth min
number in array then sum of it
and difference of it
10 marks

Q5
Find the average positive and
negative numbers upto -1
10 marks

Q6
Removing duplicates in array
10 marks

Q4 Nth max number and nth min number in array then sum of it and difference of it 10 marks

Nth max number and nth min number in array then sum of it and difference of it

Your Code

```
1 import java.util.*;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int size = sc.nextInt();
6         int[] a = new int[size];
7         for (int i = 0; i < size; i++)
8             a[i] = sc.nextInt();
9         int n = sc.nextInt();
10        Arrays.sort(a);
11        int min = a[n - 1];
12        int max = a[size - n];
13        System.out.println("Nth Minimum = " + min);
14        System.out.println("Nth Maximum = " + max);
15        System.out.println("Sum = " + (min + max));
16        System.out.println("Difference = " + (max - min));
17    }
18 }
```

Run

Save

Input

5
10 30 20 50 40
2

Output

Nth Minimum = 20
Nth Maximum = 40
Sum = 60
Difference = 20



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Q1
Palindrome number/string
10 marks

Q2
First n perfect numbers
10 marks

Q3
Decimal number equivalent to binary number and octal number
10 marks

Q4
Nth max number and nth min number in array then sum of it and difference of it
10 marks

Q5
Find the average positive and negative numbers upto -1
10 marks

Q6
Removing duplicates in array
10 marks

Q7

Q5 Find the average positive and negative numbers upto -110 marks

Find the average positive and negative numbers upto -1

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int num, psum = 0, nsum = 0, pc = 0, nc = 0;
6         while ((num = sc.nextInt()) != -1) {
7             if (num > 0) {
8                 psum += num;
9                 pc++;
10            } else {
11                nsum += num;
12                nc++;
13            }
14        }
15        if (pc > 0)
16            System.out.println("Positive Average = " + (psum / pc));
17        if (nc > 0)
18            System.out.println("Negative Average = " + (nsum / nc));
19    }
20 }
```

Input

```
10
20
-5
15
```

Output

```
Positive Average = 15
Negative Average = -7
```



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Q1
Palindrome number/string
10 marks

Q2
First n perfect numbers
10 marks

Q3
Decimal number equivalent to binary number and octal number
10 marks

Q4
Nth max number and nth min number in array then sum of it and difference of it
10 marks

Q5
Find the average positive and negative numbers upto -1
10 marks

Q6
Removing duplicates in array
10 marks

Q7

Q6 Removing duplicates in array10 marks

Removing duplicates in array

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] a = new int[n];
7         for (int i = 0; i < n; i++)
8             a[i] = sc.nextInt();
9         for (int i = 0; i < n; i++) {
10             int j;
11             for (j = 0; j < i; j++) {
12                 if (a[i] == a[j])
13                     break;
14             }
15             if (i == j)
16                 System.out.print(a[i] + " ");
17         }
18     }
19 }
```

Input

```
5
1 2 3 4 4
```

Output

```
1 2 3 4
```



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Q1
Palindrome number/string
10 marks

Q2
First n perfect numbers
10 marks

Q3
Decimal number equivalent to binary number and octal number
10 marks

Q4
Nth max number and nth min number in array then sum of it and difference of it
10 marks

Q5
Find the average positive and negative numbers upto -1
10 marks

Q6
Removing duplicates in array
10 marks

Q7
Matrix multiplication
10 marks

Matrix multiplication

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int[][] a = new int[2][2];
6         int[][] b = new int[2][2];
7         int[][] c = new int[2][2];
8         for (int i = 0; i < 2; i++)
9             for (int j = 0; j < 2; j++)
10                 a[i][j] = sc.nextInt();
11         for (int i = 0; i < 2; i++)
12             for (int j = 0; j < 2; j++)
13                 b[i][j] = sc.nextInt();
14         for (int i = 0; i < 2; i++)
15             for (int j = 0; j < 2; j++)
16                 for (int k = 0; k < 2; k++)
17                     c[i][j] += a[i][k] * b[k][j];
18         for (int i = 0; i < 2; i++) {
19             for (int j = 0; j < 2; j++)
20                 System.out.print(c[i][j] + " ");
21             System.out.println();
22         }
23     }
24 }
```

Input

```
1 2
3 4
5 6
7 8
```

Output

```
19 22
43 50
```